Highly Available Web Application with Load Balancer

You will create a **fault-tolerant web application** using:

- Multiple EC2 instances
- Application Load Balancer (ALB)
- Auto Scaling (optional but powerful)
- Health checks

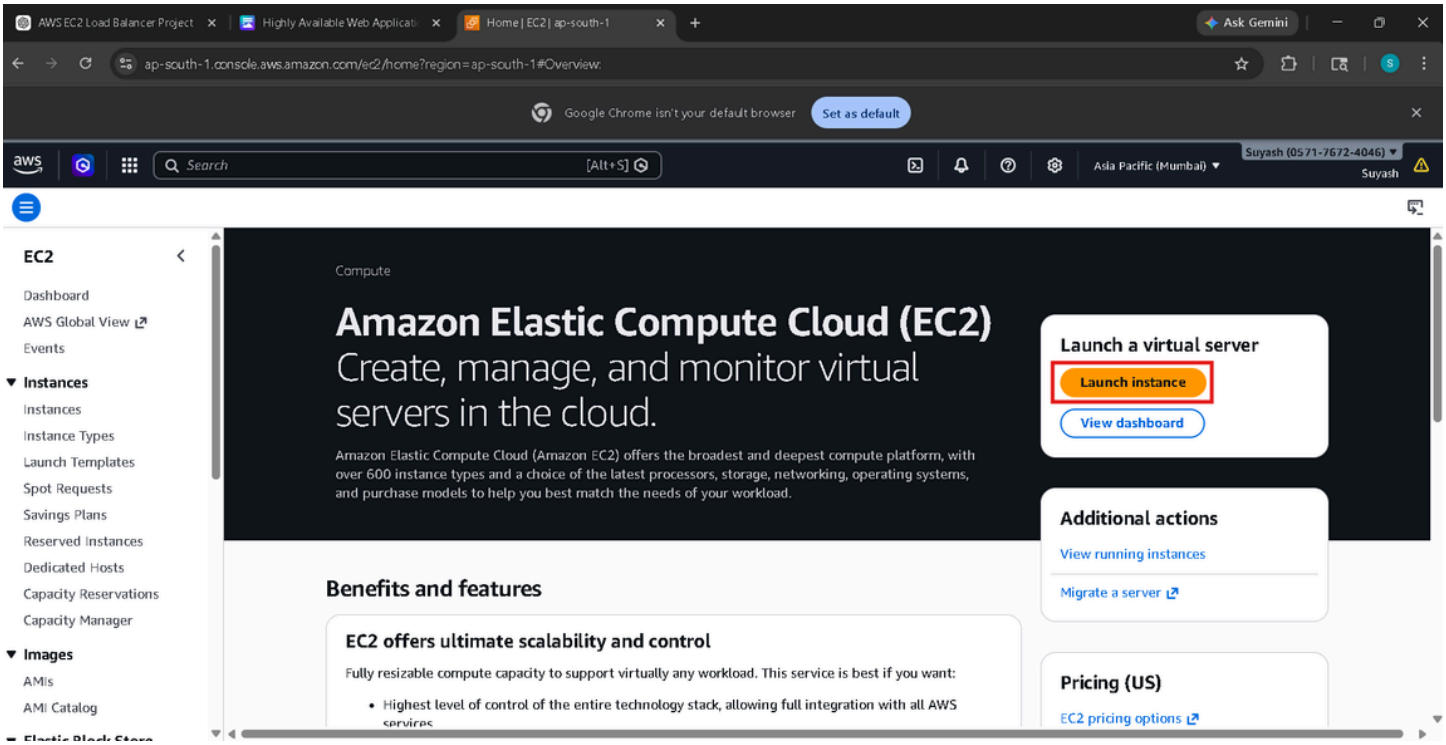
Architecture Overview

User → Load Balancer → EC2 Instance 1
EC2 Instance 2

Go on AWS Console sign in

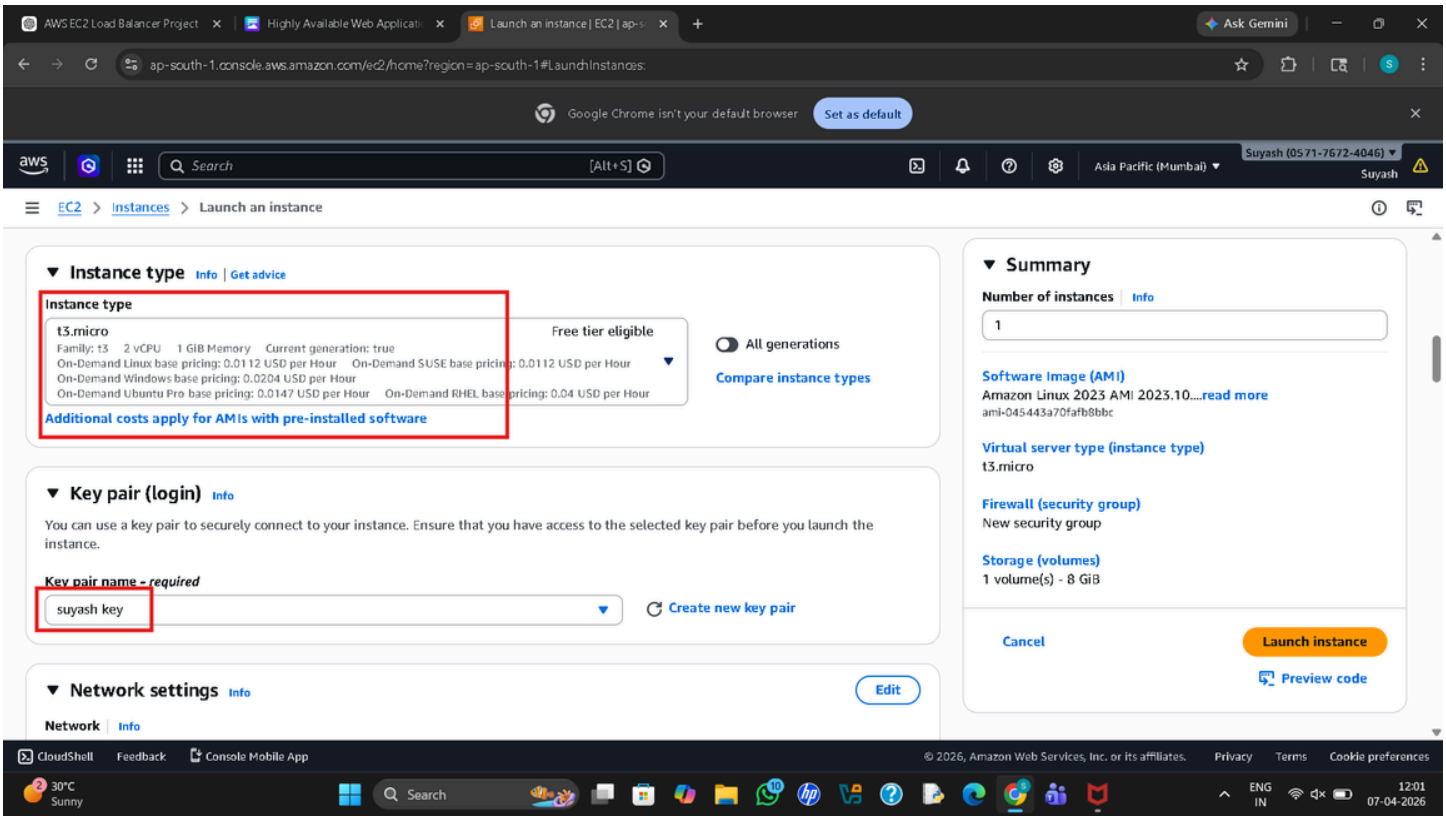
Go on ec2 Dashboard

Launch instances



Fill all details

same region and AZ



In advance Details put there shebang line

Install HTTPD

sudo yum install -y httpd

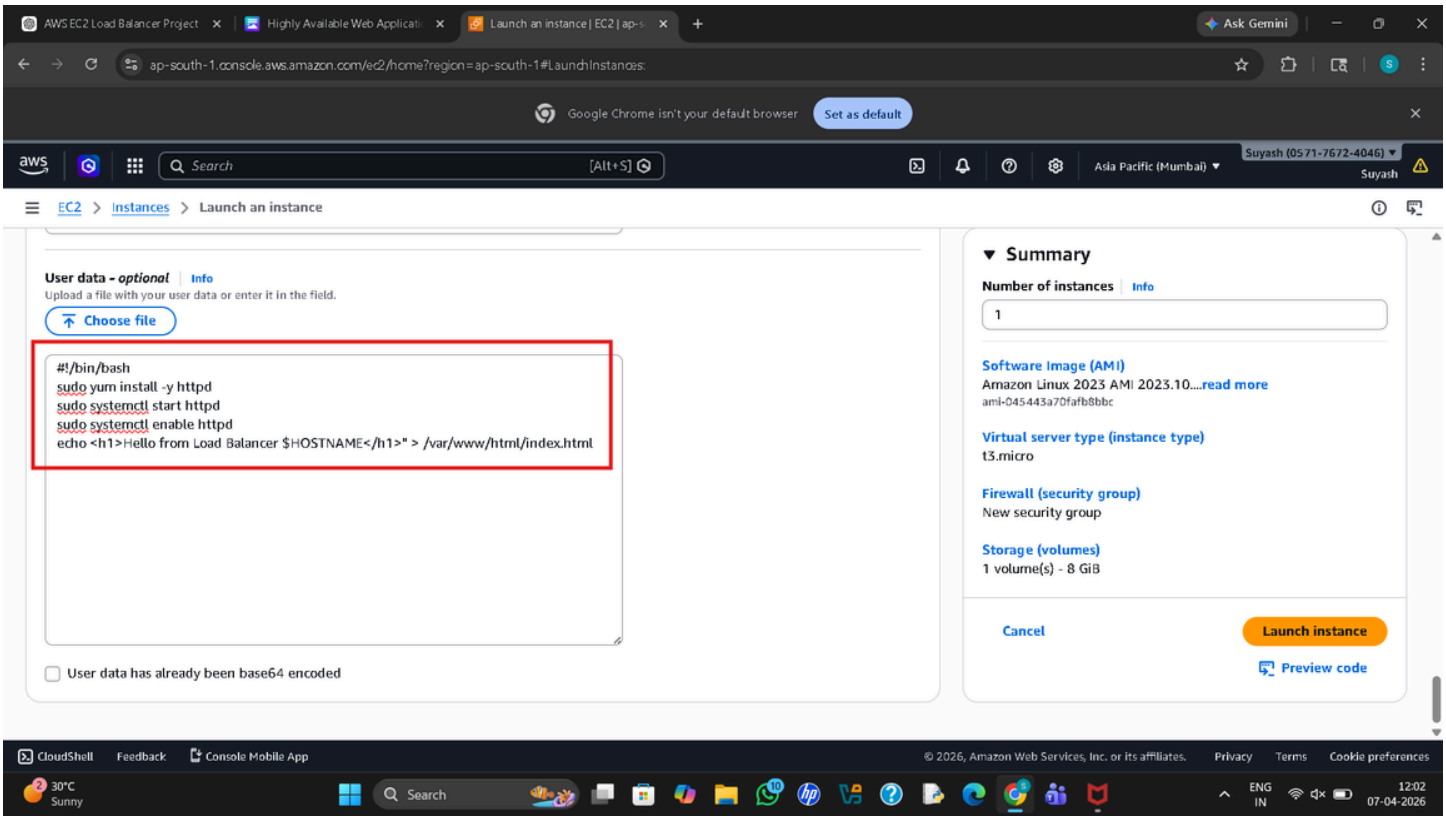
Start the server

sudo systemctl start httpd

Enable on boot

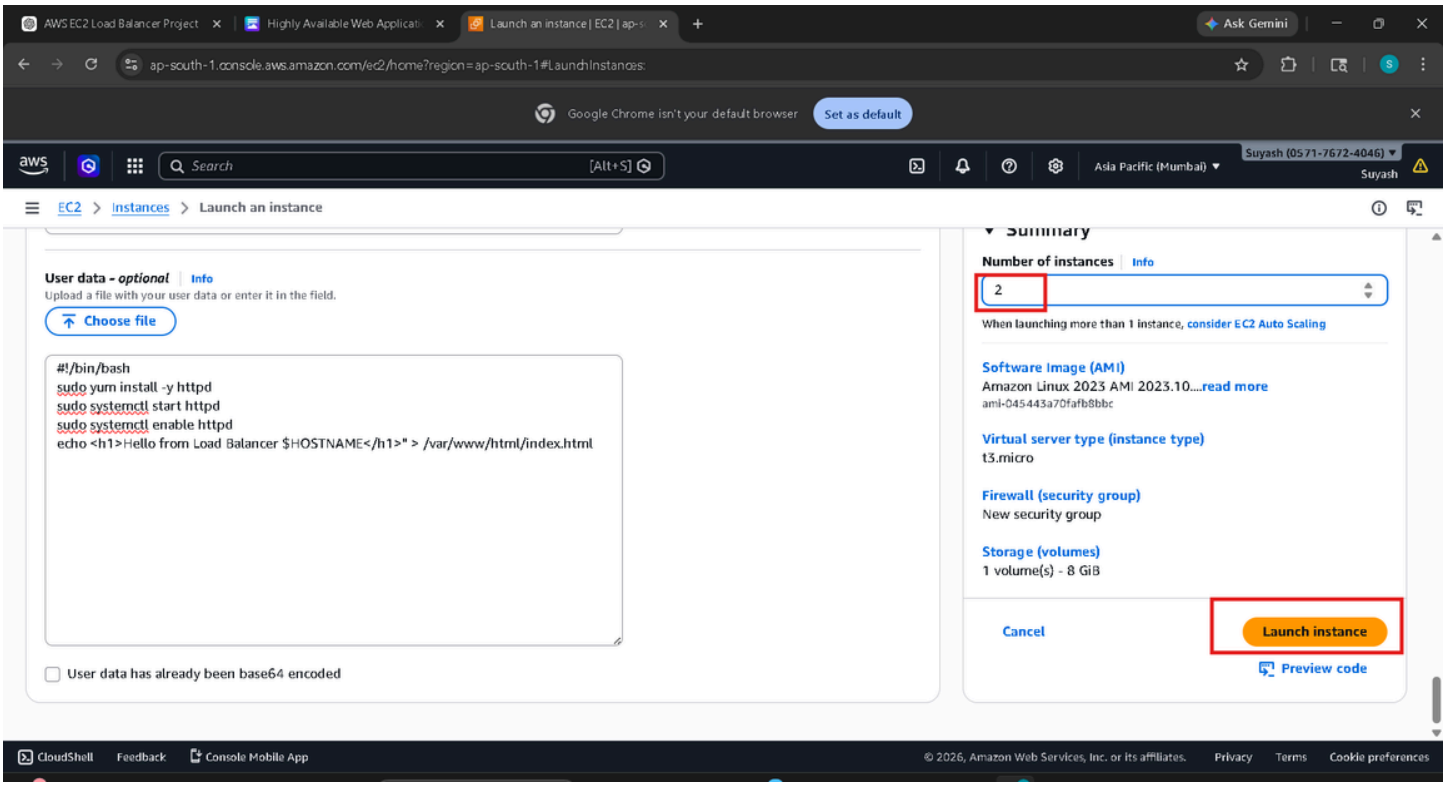
sudo systemctl enable httpdAdd custom page:

echo <h1>Hello from Load Balancer $HOSTNAME</h1>" > /var/www/html/index.html

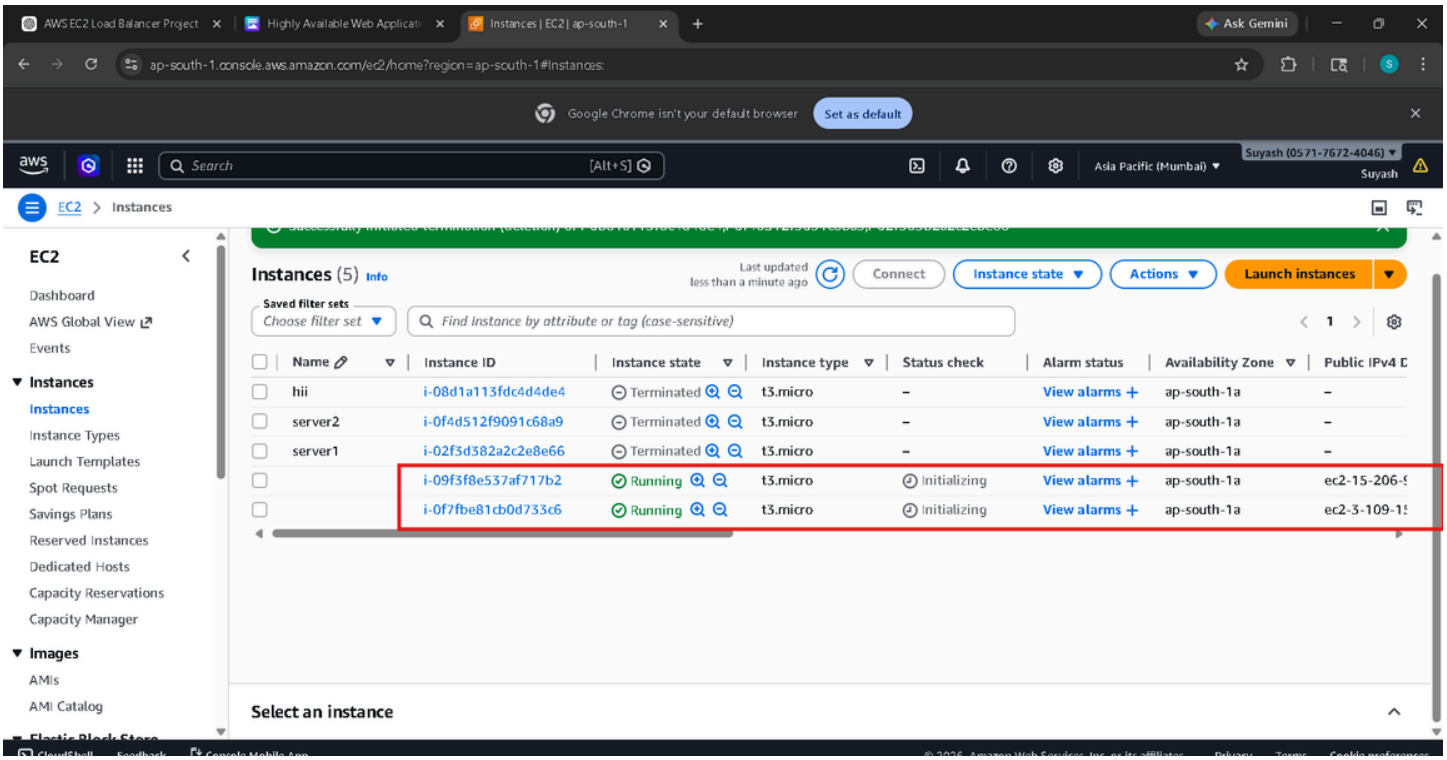


Create once 2 server

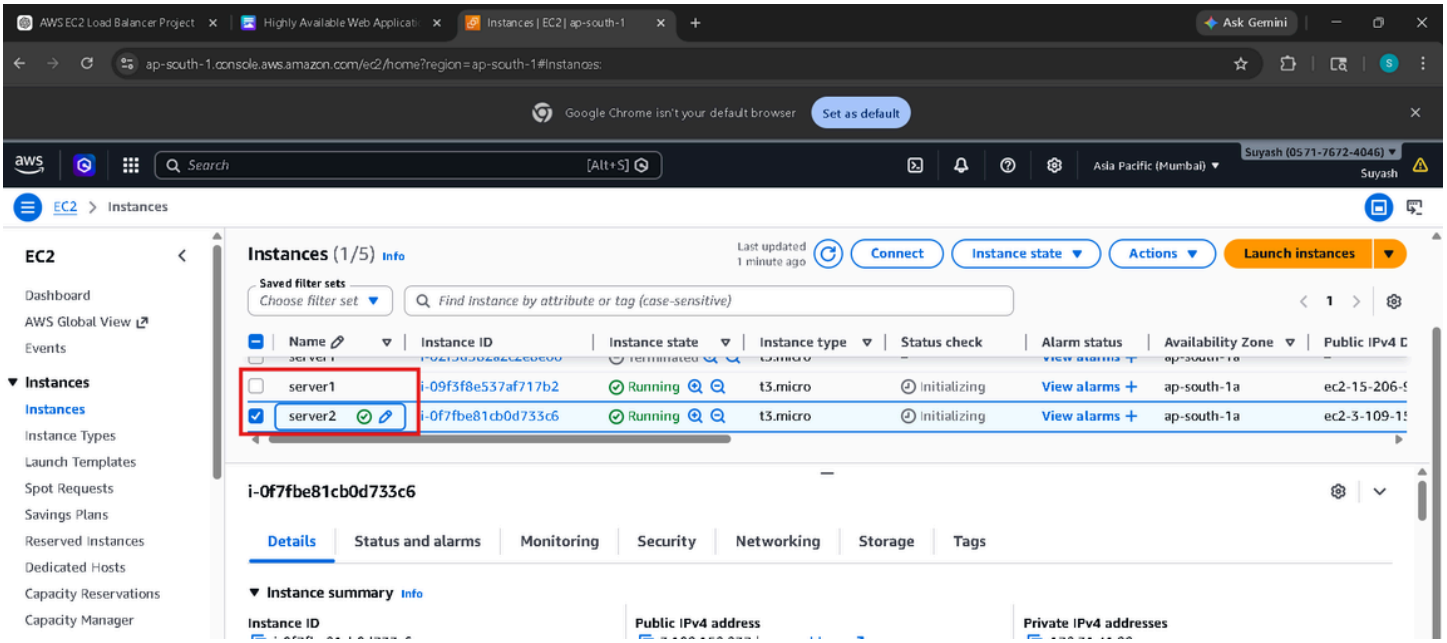
Launch instance



2 instance are running

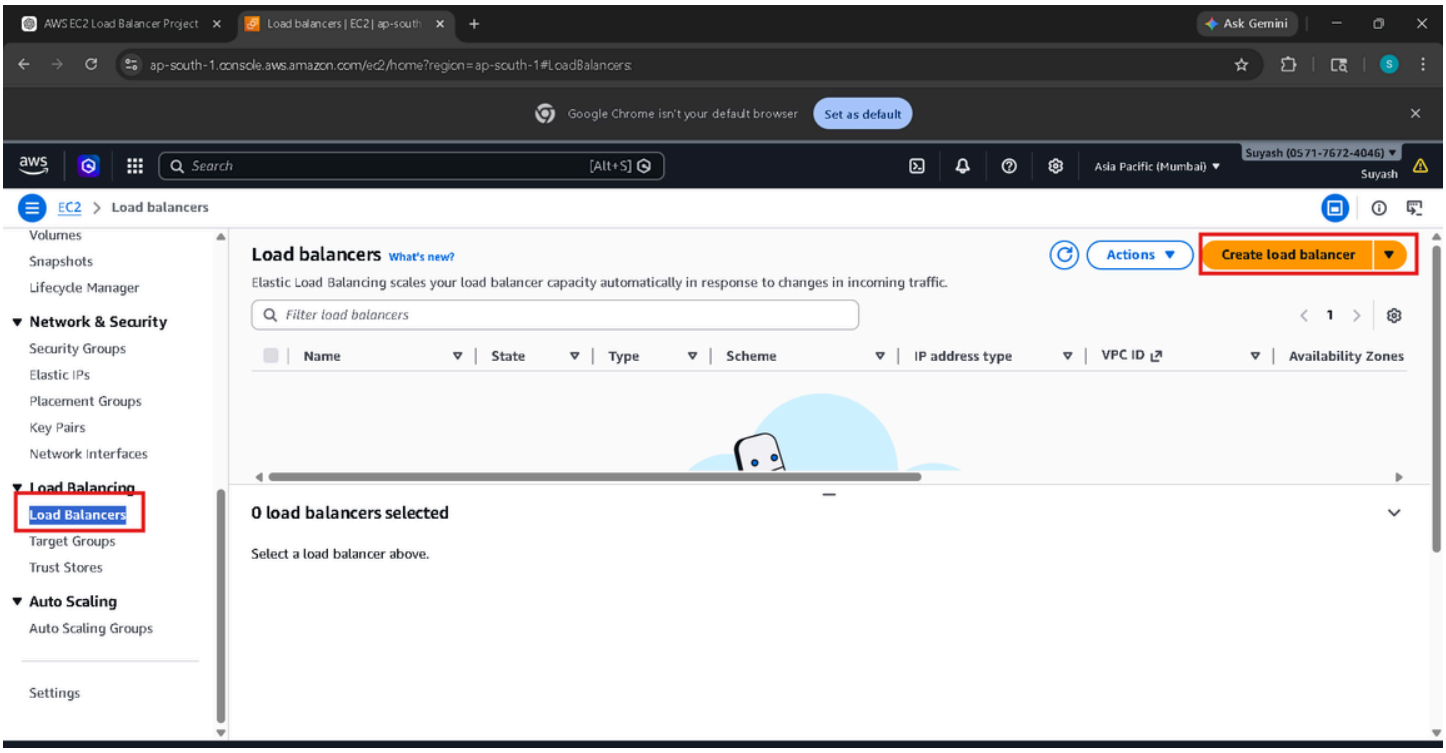


Put name as server1 and server2



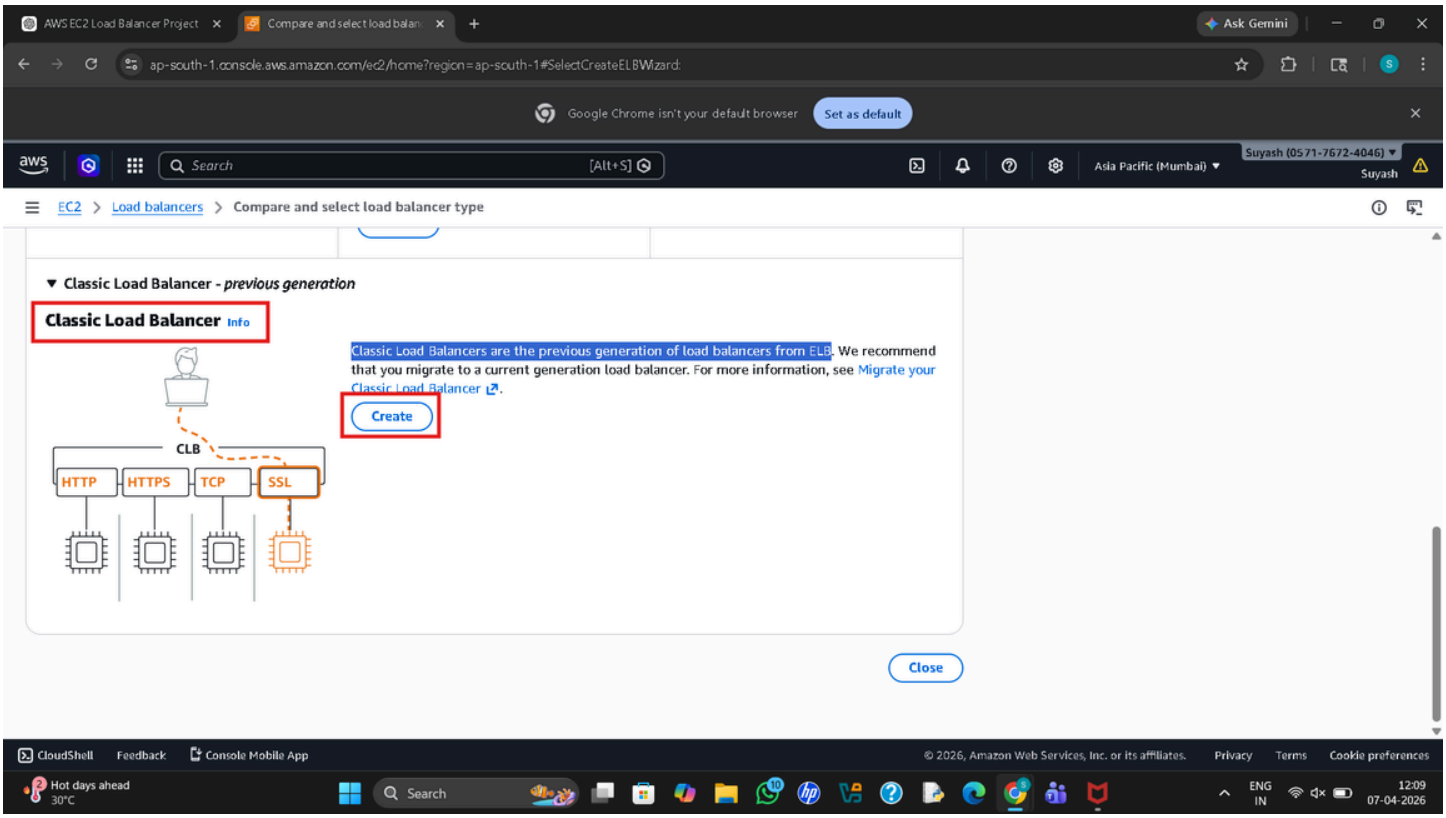
Scrool down and click on **Load Balancers**

Create load balancer



Go on **Classic Load Balancer - *previous generation***

Classic Load Balancers are the previous generation of load balancers from ELB



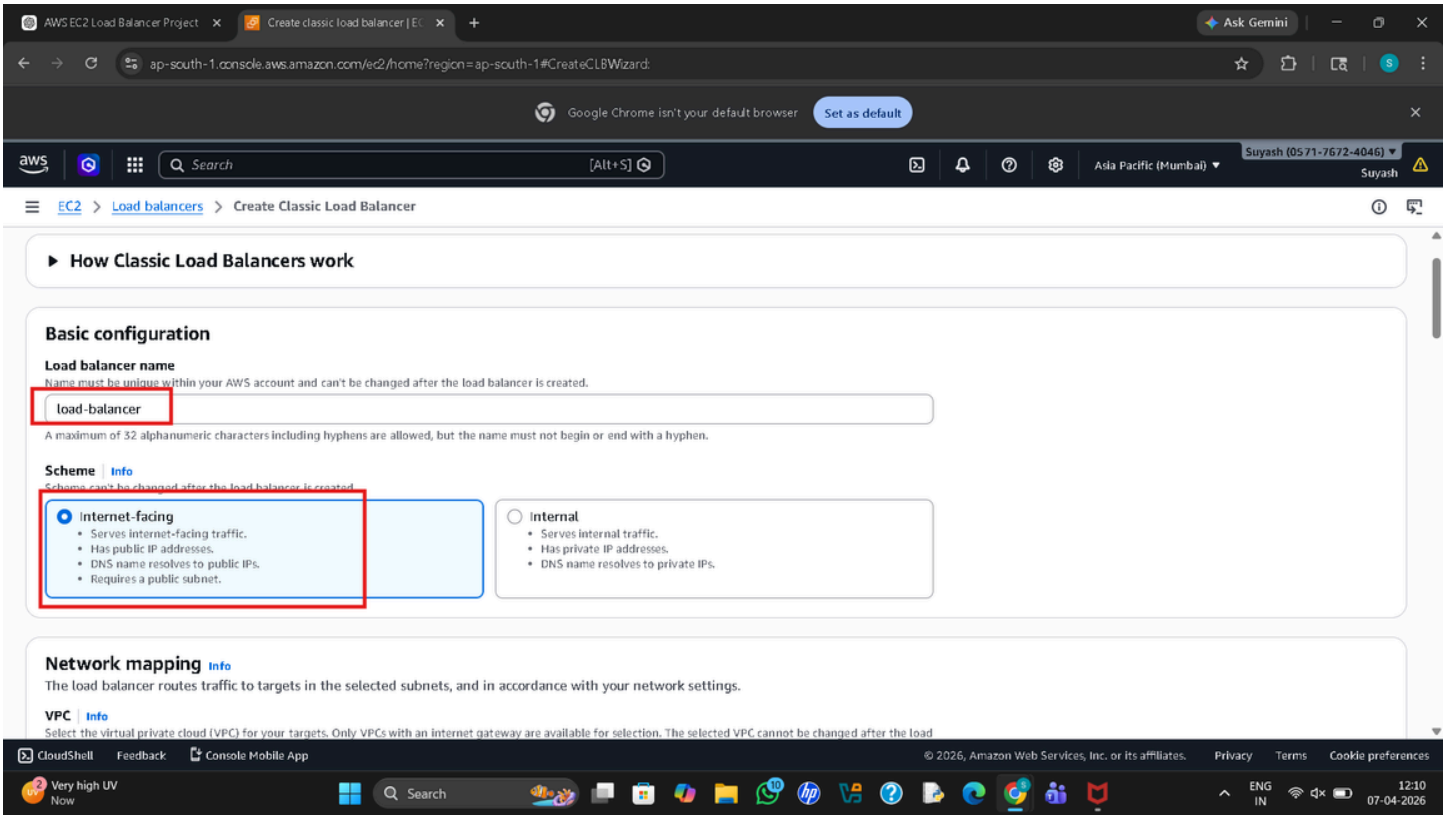
Create it Name the **Load balancer name**

load-balancer

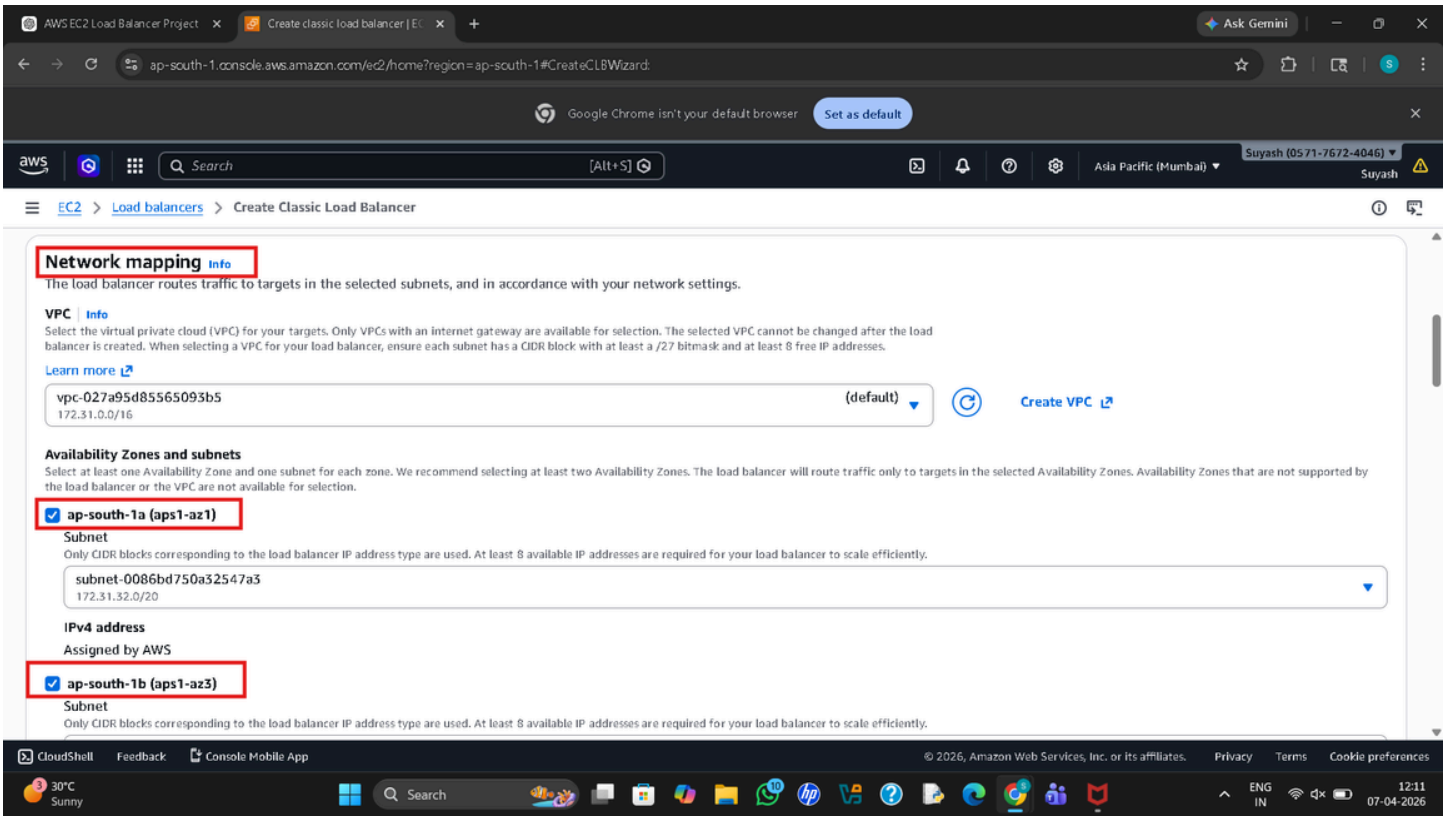
Scheme

- Internet-facing
- Serves internet-facing traffic.

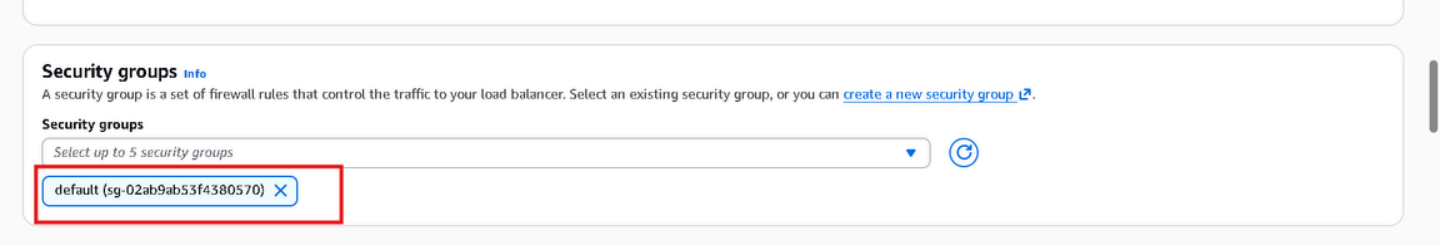
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.



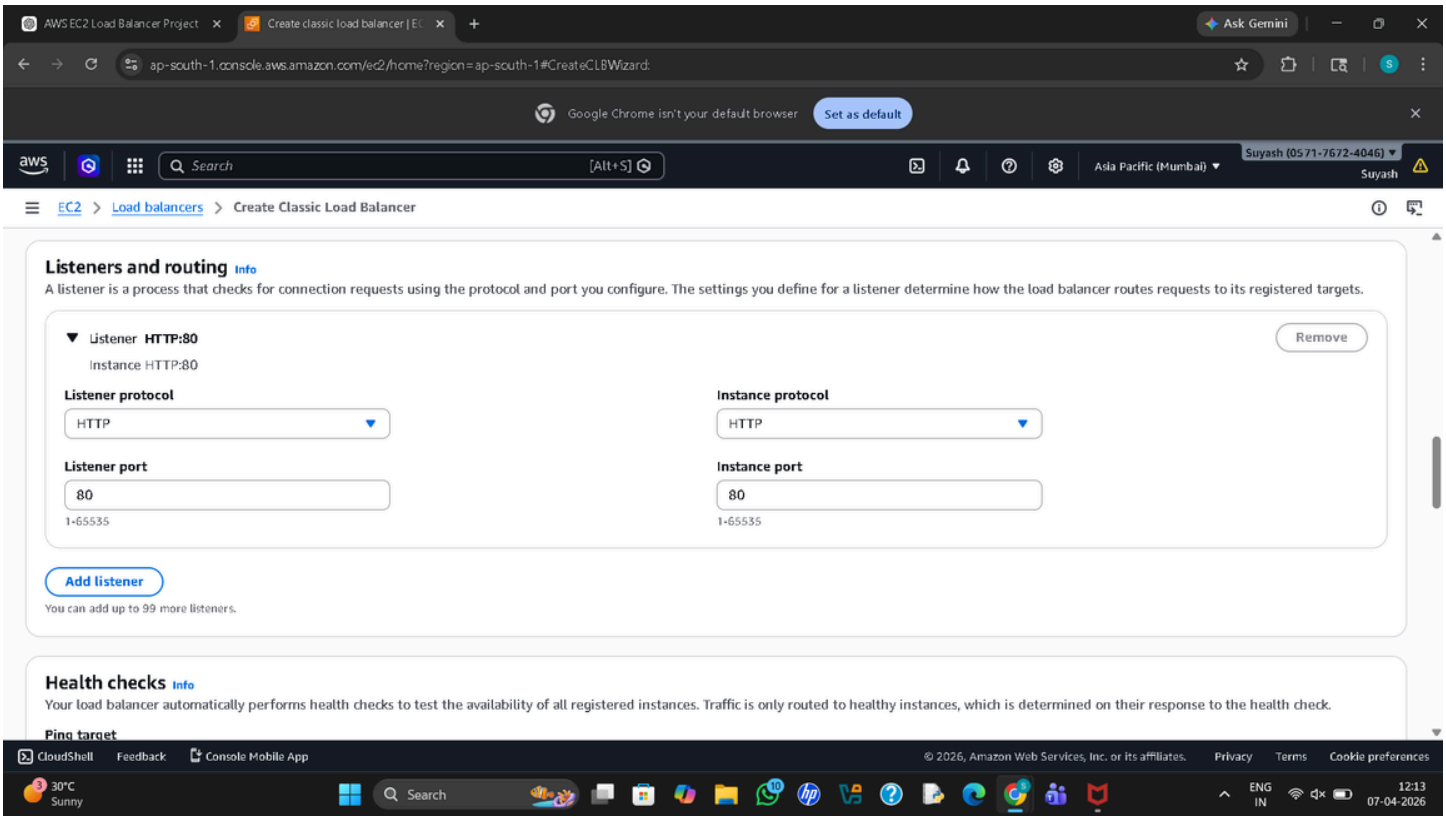
Go on Network Setting and click atleast 2 AZ



Security groups



Listeners and routing

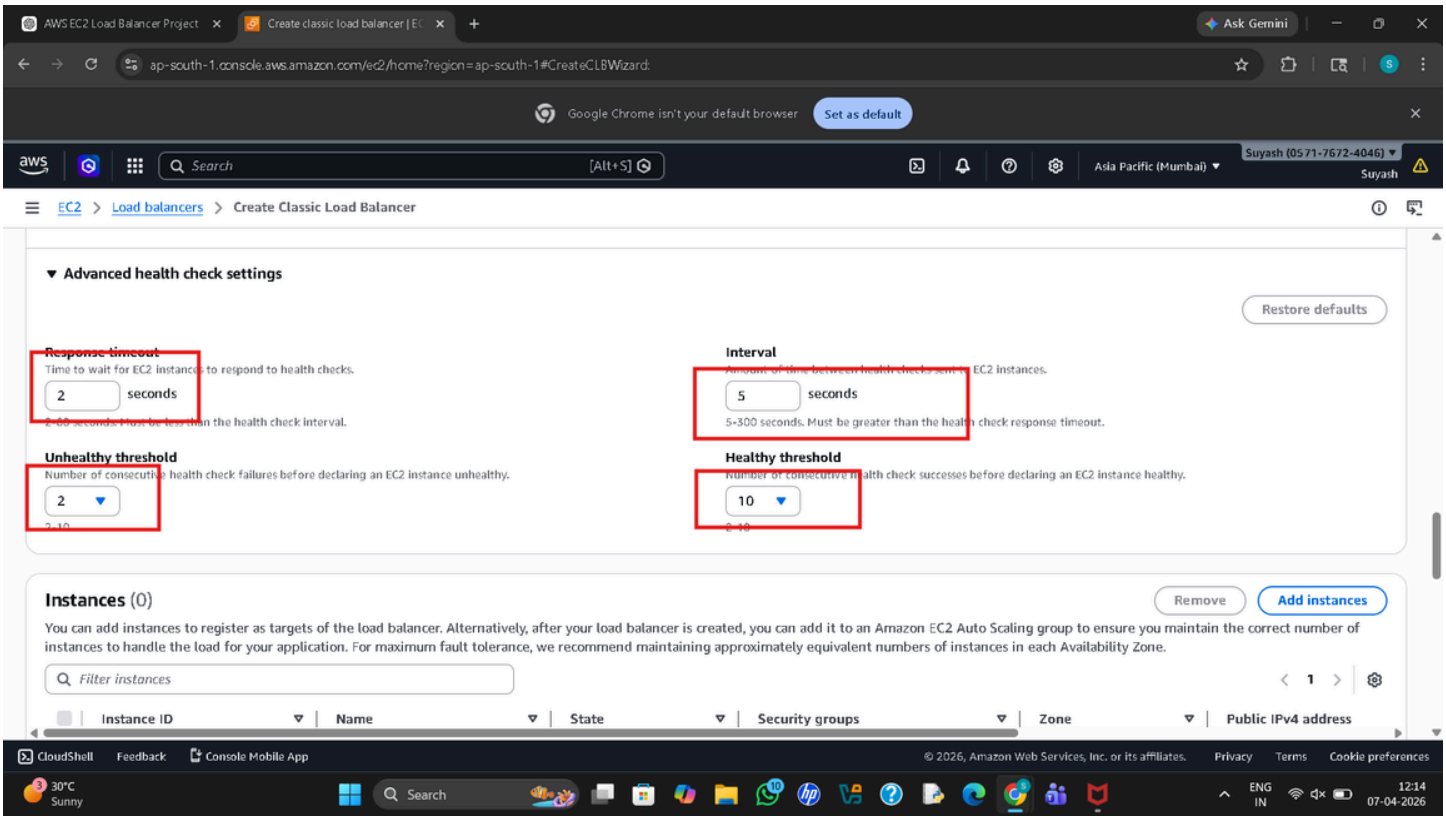


Response timeout (2s): How long LB waits for reply

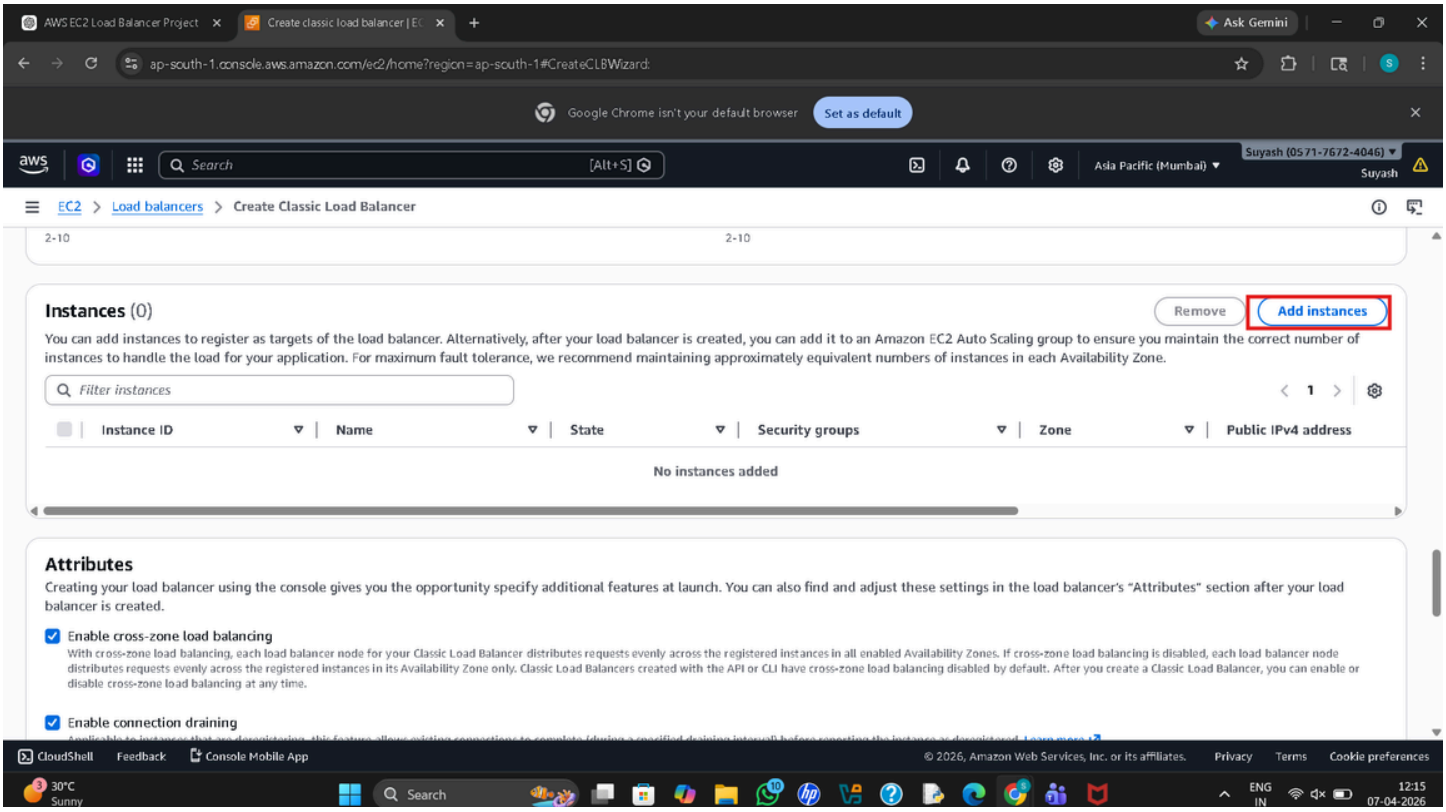
Interval (5s): How often LB checks

Unhealthy threshold (2): Fail 2 times → marked unhealthy

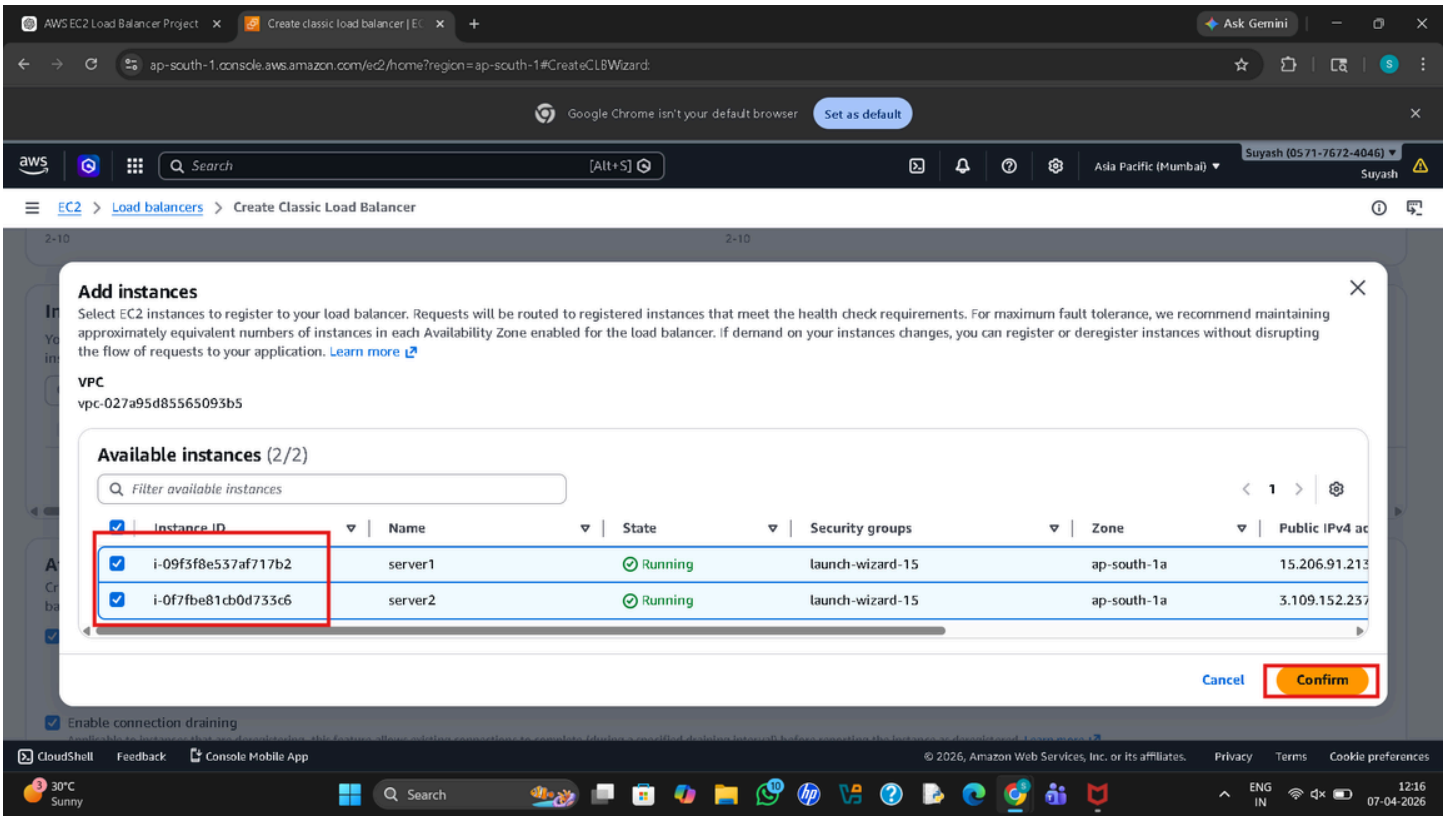
Healthy threshold (10): Pass 10 times → marked healthy



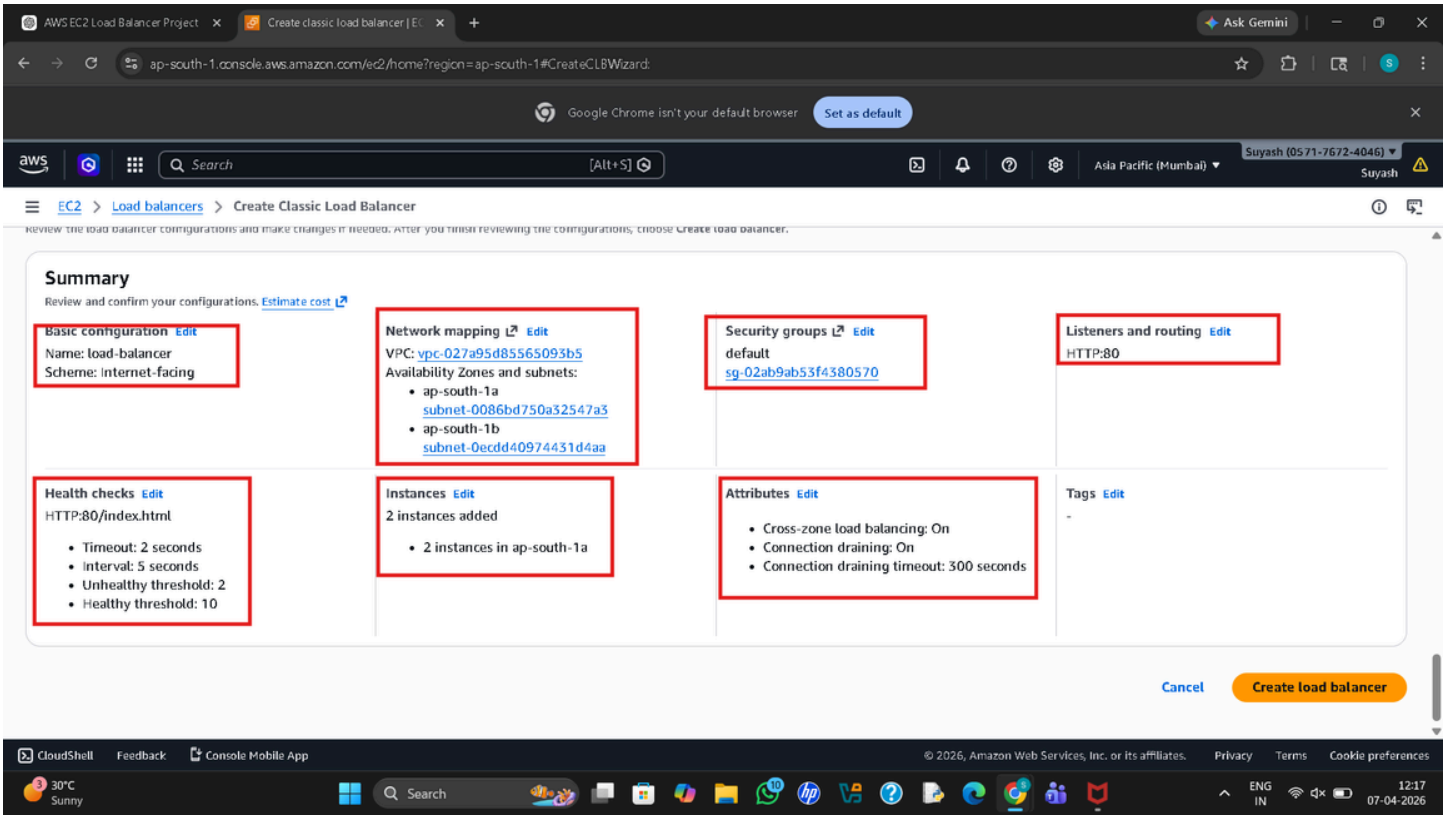
Now add your created Instance



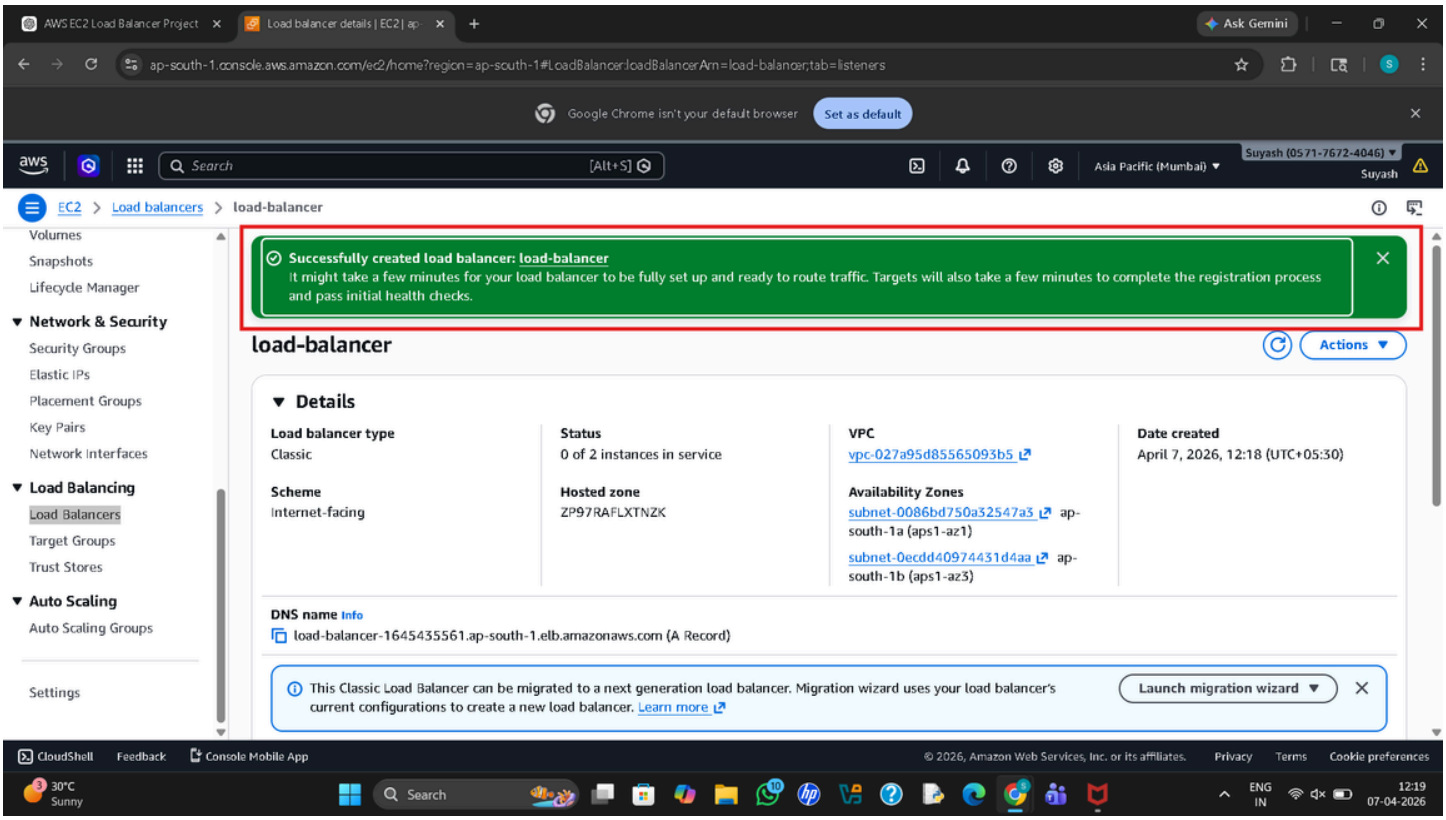
Select it and confirm 2 instance at a time



See the summary

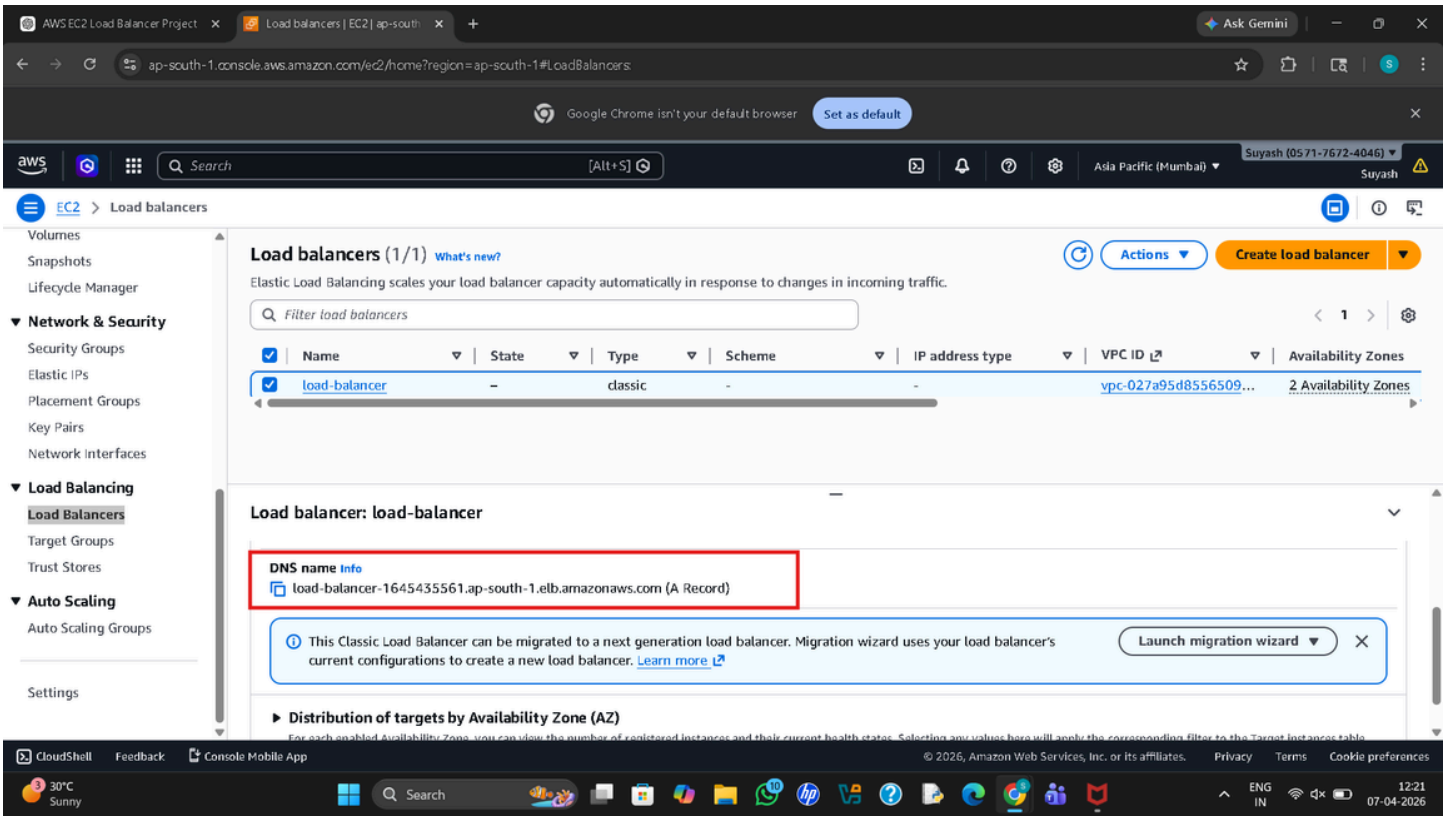


Click Create on load balancer



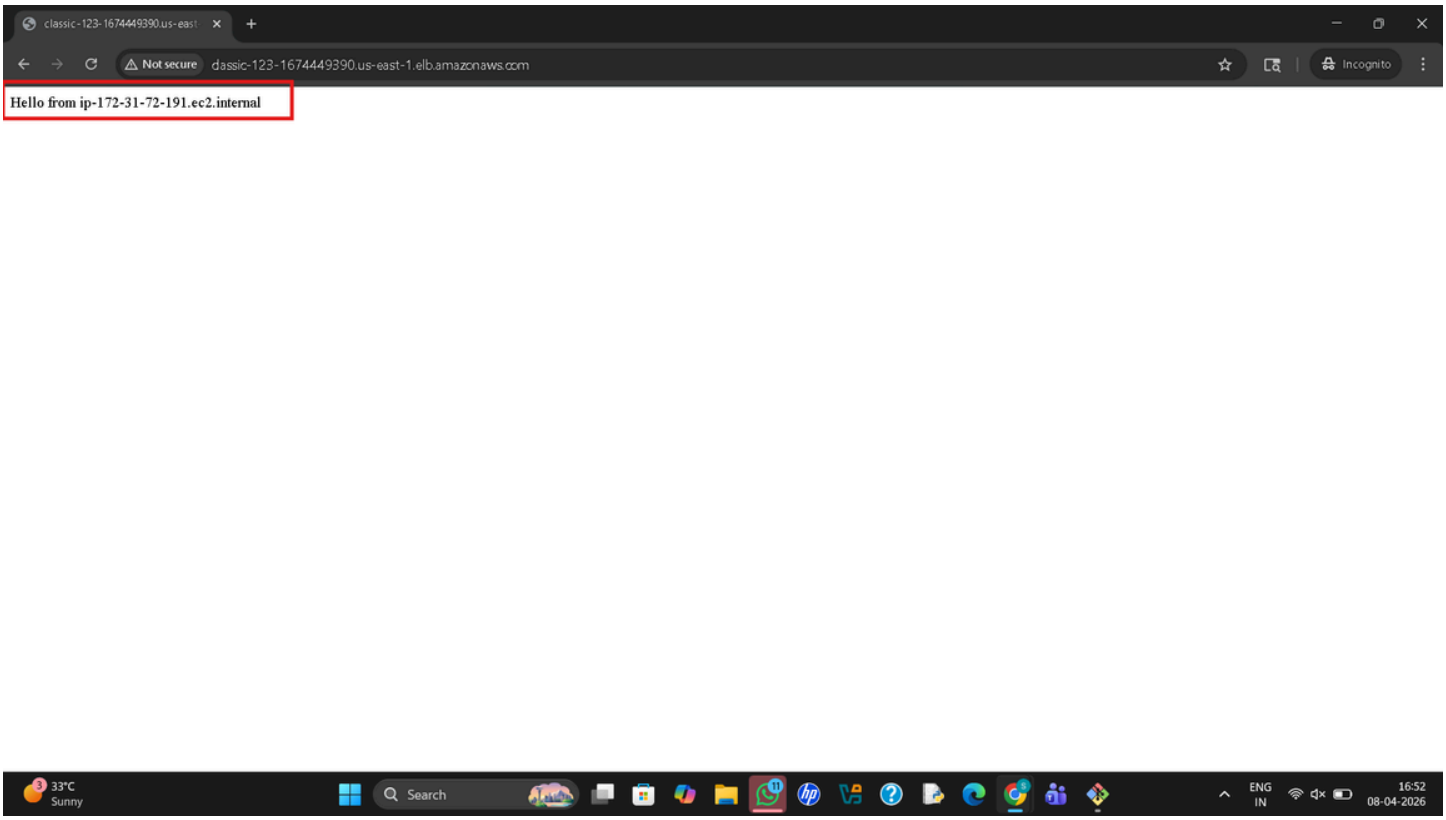
Go on load balance and click on your load balancer

you will see the DNS name copy it and paste on Browesr

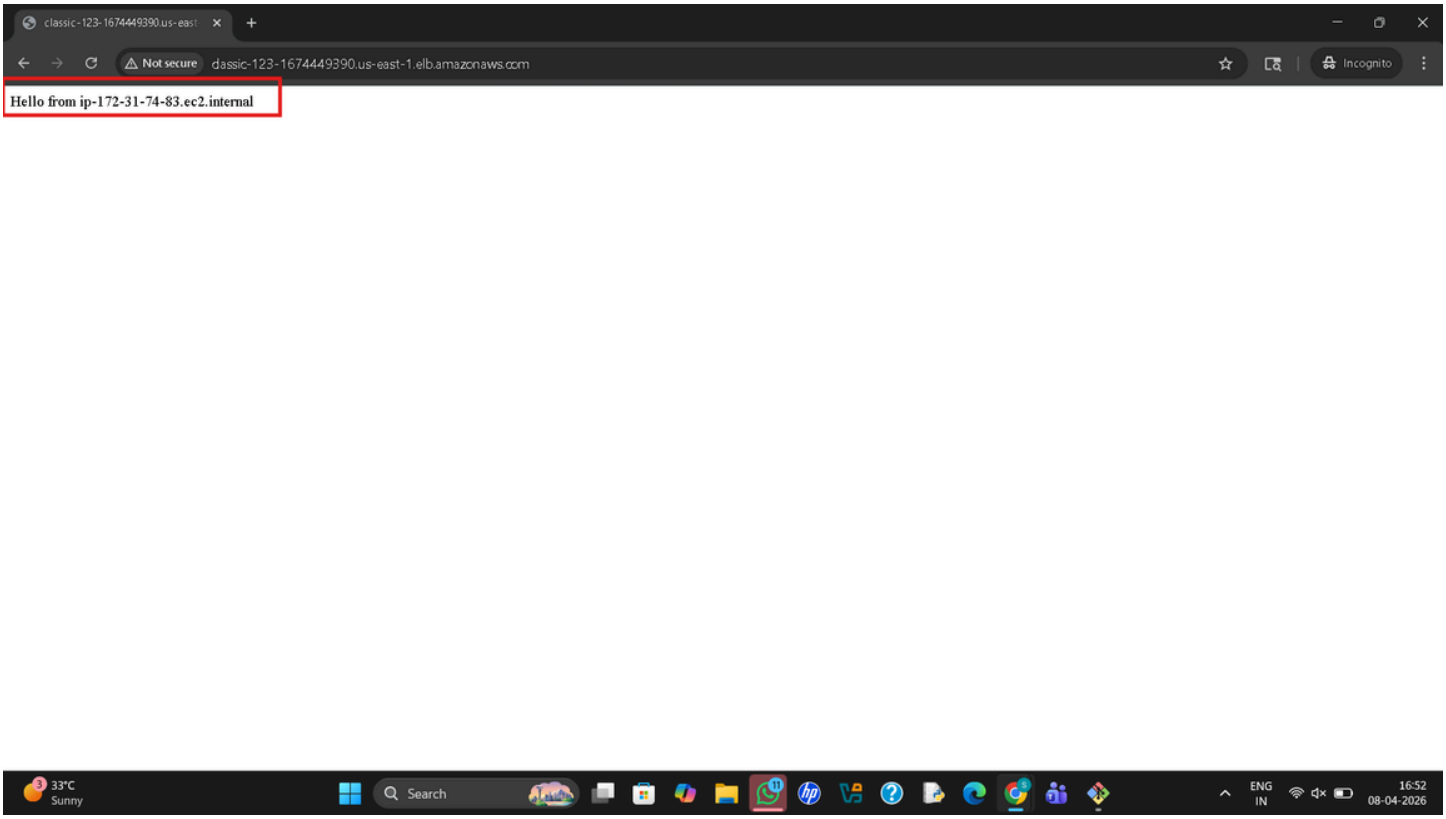


you will se the page and refresher it each time you will see new if addresses for 2 time and repeat it via refreshing.

First ip address.



Second ip address.



This project demonstrates the deployment of a highly available and fault-tolerant web application using AWS infrastructure. A Classic Load Balancer was configured to distribute incoming HTTP traffic across multiple EC2 instances deployed within the same Availability Zone. Each instance was configured with a web server (Apache) serving a static web page.

Health checks were implemented to monitor instance availability, ensuring traffic is routed only to healthy instances. Proper security group configurations were applied to allow secure

communication between the Load Balancer and EC2 instances. The setup ensures improved reliability, scalability, and fault tolerance by avoiding single points of failure.

This project highlights key concepts such as load balancing, health monitoring, high availability, and basic web server configuration in a cloud environment.